

Zheng Jiang

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Education

(2024/10-present) University of Cambridge: Mathematical Tripos, first year score: 82/100, ranking: 26/259, second year score: 68/100, ranking: 97/248.

Awards and Honors

(2024/11) Scholar at Selwyn College

(2024/11) Tripos Prize and Ratcliffe Prize Obtained first class for part IA tripos

(2024/06) Alibaba Global Mathematics Competition Ranked 274/50000+ Globally in the first round, qualified for the final round.

(2024/02) UK Integration Bee First round ranked 6/42 in Cambridge. Second round ranked 6/23 in UK

(2024/12) ICMC (Imperial Cambridge math competition) Qualified for the final round.

(2023/06) STEP (Sixth Term Examination Paper) STEP 2: S (98/120), STEP 3: S (102/120)

(2022/11) Physics Unlimited Premier Competition (held by Princeton University) Ranked 34/500 globally, school champion.

(2022/11) Euclid Math Competition (held by the University of Waterloo) Top 0.6% globally, school champion.

Projects

(2025/08-2025/10) Independent Research:

Self-studied Introduction to Smooth Manifolds (Lee, GTM) and Lectures on Differential Geometry (Chern) as theoretical background. Deep reading of Hatcher's The Kirby Torus Trick for Surfaces and Milnor's Morse Theory; extracted and reorganised key arguments for a student audience. Wrote an expository essay clarifying the relationship between handle decompositions, Morse functions, and smooth structures in low dimensions. Developed several original proofs of intermediate results (e.g. existence of a chain-like atlas, fundamental theorems in Morse theory, handle smoothing theorems).

(2024/11-2025/02) Computational Projects: Conducted numerical analysis of ODEs using various approximation methods, evaluating and comparing their performance. Applied the shooting method to construct eigenfunctions and determine eigenvalues for specific Sturm-Liouville problems, interpreting the results physically through first principles. Analyzed the row-reduced echelon form of matrices in finite fields and examined the relationships between rank, annihilators, and kernel, contrasting these findings with matrices of real or complex entries.

(2020-present) Desmos Math Visualization Projects: Utilized functions and equations to create artwork, incorporating linear algebra to develop a 3D visual effect rendering engine for realistic animations. Visualized mathematical concepts to enhance understanding for math learners. Shared these projects on Reddit and academic platforms, receiving positive feedback from the community and being used informally as teaching aids by other students.

Experiences

(2023/07-present) Private Tutor (100h) Provide one-to-one tutoring in A-Level, IB Mathematics and STEP exams, designing personalised notes, problem sets, and revision plans. Conduct Oxbridge-style mock interviews in Mathematics, training students to explain their reasoning clearly under time pressure.

(2024/07-2024/09) Topology Tutoring (40h) Tutored two University of Leeds students in pointset topology (quotient spaces, connectedness, compactness), designing problem sets and revision plans. Both students passed their resit examinations.

Skills & Hobbies

Python, LaTeX, MATLAB

Painting, Sketching, Drawing, Calligraphy, Traditional Chinese Painting, Frisbee, MOBA-like games

Languages

Chinese (native), English (fluent; TOEFL iBT 110/120)

(Any references are available upon request)